

CSE 208

Online on SSSP

Duration: 30 minutes

Problem : Nearest Emergency Center

A country has n cities connected by m one-way roads. Each road has a positive travel time. Some cities contain emergency centers.

For every city, determine the minimum time required to reach any emergency center by following the direction of the roads.

If a city itself has an emergency center, the minimum time is 0.

If a city cannot reach any emergency center, print -1.

Input Format

The first line contains three integers:

n m k

where:

- n = number of cities
- m = number of one-way roads
- k = number of emergency centers

The second line contains k integers, representing the cities that contain emergency centers.

The next m lines each contain three integers:

u v w

meaning there is a one-way road from city u to city v with travel time w .

Output Format

Print n lines.

For each city from 1 to n, print the minimum time required to reach any emergency center.

If no emergency center is reachable from that city, print:

-1

Constraints

$$1 \leq n \leq 10^5$$

$$1 \leq m \leq 2 \times 10^5$$

$$1 \leq k \leq n$$

$$1 \leq u, v \leq n$$

$$1 \leq w \leq 10^9$$

Sample Input 1

7 7 2

1 5

2 1 4

3 1 2

2 3 1

4 2 5

4 3 8

6 5 3

7 6 2

Sample Output 1

0

3

2

8

0

3

5

Explanation

There are emergency centers in cities 1 and 5.

From city 2, the shortest directed path to an emergency center is:

2 -> 3 -> 1

with total time:

$$1 + 2 = 3$$

From city 7, the shortest directed path to an emergency center is:

7 -> 6 -> 5

with total time:

$$2 + 3 = 5$$

Sample Input 2

6 4 2

1 6

1 2 3

2 3 4

5 4 2

6 5 2

Sample Output 2

0

-1

-1

-1

-1

0